# Electric Vehicle Population Data — Data Cleaning & Analysis Practical

### Instructions to students

Use the provided **Electric Vehicle Population Data** dataset  ( around 70 mb data ).

Your goal is to inspect, clean, validate, and export the dataset.

For every cleaning operation, you must be able to explain:

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## PART A — IMPORTING & UNDERSTANDING THE DATA

### Task 1 — Import the dataset

Import the provided dataset into Pandas using the appropriate function for its file format.

After importing, display the first few records.

**Questions:**

1. What function did you use to import the data?
2. How many rows and columns does the dataset contain?
3. What does one row represent?
4. What is the purpose of the main columns in the dataset?

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### Task 2 — Inspect the dataset structure

Display the column names and data types.

Classify the columns into:

- Categorical
- Numerical
- Date/time
- Identifier/code

**Question:**

Which columns do you think require cleaning before numerical analysis? Explain why.

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### Task 3 — Understand the dataset before cleaning

Answer the following using the raw dataset:

1. Which column identifies the vehicle manufacturer?
2. Which column identifies the vehicle model?
3. Which column represents the type of electric vehicle?
4. Which columns describe location?
5. Which columns contain information related to electric range or vehicle price?
6. Which column represents the model year?

Do not modify the dataset yet.

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# PART B — MISSING VALUES

### Task 4 — Find missing values

Calculate the number of missing values in every column.

Then answer:

1. Which columns contain missing values?
2. Which column has the highest number of missing values?
3. Which columns have very few missing values?
4. Are there any columns with almost no missing data?

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### Task 5 — Calculate missing-value percentages

Calculate the percentage of missing values for every column.

Create a simple conclusion:

> "The column with the highest percentage of missing data is _____."

Then explain whether that amount of missing data appears:

- Low
- Moderate
- High

Do not remove anything yet.

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### Task 6 — Decide how to handle missing values

For every important column containing missing values, decide whether you should:

- Drop rows
- Drop the column
- Fill using median
- Fill using mode

You must give a reason for each decision.

**Important:** Do not automatically use the same technique for every column.

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### Task 7 — Handle numerical missing values

Identify the numerical columns that contain missing values.

Fill appropriate numerical missing values using the method covered in the guide.

After cleaning:

1. Check the missing-value count again.
2. Compare the missing-value count before and after cleaning.
3. Explain what changed.

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### Task 8 — Handle categorical missing values

Identify categorical columns containing missing values.

If an appropriate categorical column has missing values, handle them using the technique from the guide.

Then verify that the operation worked.

# PART C — DATA INCONSISTENCY

## Task 9 — Check text consistency

Inspect text-based columns such as:

- Make
- Model
- City
- County
- State

Look for possible problems such as:

- Leading/trailing spaces
- Different capitalization
- Inconsistent text representation

Choose **at least two applicable columns** and standardize them using the technique from the guide.

Then explain:

> What type of inconsistency were you trying to remove?

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## Task 10 — Investigate categorical values

For important categorical columns, inspect the unique values.

For example, investigate the vehicle type column.

Answer:

1. What categories are present?
2. Are the categories consistently written?
3. Are there obvious duplicate representations of the same category?
4. Did you find anything that needed standardization?

Do not replace values unless you have identified a genuine inconsistency.

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## Task 11 — Type conversion

Inspect the numerical columns.

Identify columns that have been imported as text even though they represent numbers.

Convert the applicable columns into proper numerical data types using the technique from the guide.

After conversion:

Before → data type
After → data type

Explain why this conversion is necessary.

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# PART D — DUPLICATES

### Task 12 — Identify duplicate rows

Check the dataset for completely duplicated rows.

Answer:

1. How many duplicate rows exist?
2. What percentage of the dataset is duplicated?
3. Does the dataset actually require duplicate removal?

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### Task 13 — Remove exact duplicates

If duplicate rows exist, remove them.

If there are no duplicate rows:

> State clearly that no exact duplicate records were found and therefore no records were removed.

This is important: **do not create duplicates just to demonstrate the function.**

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### Task 14 — Investigate possible vehicle duplicates

Identify the column(s) that could potentially act as a vehicle identifier.

Investigate whether the same identifier appears more than once.

Then answer:

> Does the presence of a repeated identifier necessarily mean the record is an error?

Explain your reasoning.

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# PART E — OUTLIERS

## Task 15 — Identify suitable numerical columns

Before finding outliers, choose numerical columns where outlier detection makes sense.

Select at least **two numerical columns**.

For each selected column, explain:

> Why might an unusually high or low value be an outlier?

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## Task 16 — Detect outliers using IQR

For one suitable numerical column:

1. Calculate Q1.
2. Calculate Q3.
3. Calculate IQR.
4. Calculate the lower bound.
5. Calculate the upper bound.
6. Identify values outside these limits.
7. Count the number of outliers.

Do not simply delete them.

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## Task 17 — Investigate the outliers

Look at the records identified as outliers.

Answer:

1. What are some of the extreme values?
2. Which vehicle models/manufacturers appear among them?
3. Do these values look impossible or merely unusual?
4. Could an extreme value represent a legitimate vehicle?

Write a short conclusion.

This is an important data-science step:

> **An outlier is not automatically an error.**

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### Task 18 — Apply the Z-score method

Using another suitable numerical column:

1. Calculate the mean.
2. Calculate the standard deviation.
3. Calculate the Z-score.
4. Identify records with an absolute Z-score greater than 3.
5. Count them.

Compare the results with the IQR method.

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### Task 19 — Compare IQR and Z-score

Answer:

> Did the IQR method and Z-score method identify the same records?

If not:

> Why do you think the results are different?

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# PART F — OUTLIER HANDLING

### Task 20 — Decide how to handle the outliers

For one selected numerical column, decide whether to:

- Keep the outliers
- Remove the outliers
- Cap the outliers

You must justify your decision based on the meaning of the data.

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### Task 21 — Apply outlier capping

Using the method from the guide, cap the extreme values using the calculated lower and upper limits.

Create a new column rather than destroying the original data.

For example:

Original column
↓
Capped column

Then compare some original and capped values.

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### Task 22 — Apply log transformation

For an appropriate numerical column with a strongly skewed distribution, apply the log transformation from the guide.

Answer:

> What is the purpose of applying a log transformation to a highly skewed numerical variable?

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# PART G — BASIC DATA ANALYSIS AFTER CLEANING

Now that the dataset is cleaned, perform some simple analysis.

### Task 23 — Most common manufacturers

Determine the manufacturers with the highest number of vehicles in the dataset.

Answer:

> Which manufacturer appears most frequently?

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### Task 24 — Most common vehicle models

Find the most frequently occurring vehicle models.

Answer:

> Which models appear most frequently?

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## Task 25 — BEV vs PHEV

Compare the two main electric-vehicle types.

Answer:

1. How many BEVs are present?
2. How many PHEVs are present?
3. Which type is more common?

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## Task 26 — Model-year analysis

Investigate the model-year column.

Answer:

1. What is the earliest model year?
2. What is the latest model year?
3. Which model years occur most frequently?
4. Are there any unusual model years that should be investigated?

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## Task 27 — Electric range analysis

Analyze the electric-range column.

Answer:

1. What is the average electric range?
2. What is the minimum?
3. What is the maximum?
4. Are there potential outliers?
5. What can you conclude about the range values?

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## Task 28 — Geographic analysis

Analyze the location columns.

Answer:

1. Which counties contain the most registered EVs?
2. Which cities appear most frequently?
3. Which state values are present?
4. Are there any suspicious or inconsistent location values?

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# PART H — FINAL VALIDATION

### Task 29 — Validate the cleaned dataset

After completing all cleaning operations, check:

1. Remaining missing values
2. Remaining duplicate rows
3. Numerical data types
4. Text consistency
5. Outlier treatment

Write a short statement:

> "The dataset is ready for analysis because _____."

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### Task 30 — Export the cleaned dataset

Export the final cleaned DataFrame as:

cleaned_ev_data.csv

Do not include the Pandas index.

Then create a compressed version using the compression technique from the guide.

# Final challenge — Think like a Data Scientist

### Task 31 — Write a cleaning report

Write **5–8 sentences** explaining:

1. What problems you found in the raw dataset.
2. Which missing-value techniques you used.
3. Which inconsistencies you corrected.
4. Whether duplicates existed.
5. How you handled outliers.
6. What changed after cleaning.
7. Whether you consider the final dataset ready for analysis.